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Phosphorus doped nanocrystalline diamond for supercapacitor application

Siyu Yu,[a,b] Jing Xu,[a] Hiromitsu Kato,[c] Nianjun Yang,*[a] Anna Schulte,[d] Holger Schönherr,[d] and Xin Jiang*[a]

Dedication ((optional))

Abstract: Heavily phosphorus-doped nanocrystalline diamond (P-NCD) has been grown using a plasma enhanced chemical vapor deposition technique and further applied as an electrode for the construction of supercapacitors. This P-NCD electrode shows a capacitance of 11.40 μF cm^{-2} in 1.0 M Na_2SO_4 at a scan rate of 10 mV s^{-1} and behaves as a n-type semiconductor electrode in redox-active electrolyte of 0.05 M $Fe(CN)_6^{3-/4-}$ + 1.0 M Na_2SO_4. The post-thermal treatment of as-grown P-NCD films in vacuum at high temperatures for hours leads to the achievement of much higher capacitances. At the scan rates of 10 mV s^{-1} and 20 mV s^{-1}, the capacitances are up to 2.01 mF cm^{-2} and 63.56 mF cm^{-2} for an electrical double layer capacitor and a pseudocapacitor, respectively. Such high capacitances originate from the improved electrical conductivity, varied surface state and surface functional groups, and changed content of non-carbon diamond inside the P-NCD films during annealing treatment. Therefore, P-NCD films are quite promising as the electrode material for supercapacitor applications.

Introduction

Supercapacitors (SCs) or electrochemical capacitors have been recognized as one of the most promising electrochemical energy storage devices.[1,2] They fill the gap between conventional capacitors and rechargeable batteries. In comparison to batteries, SCs feature higher power densities, faster charging/discharging rates, wider ranges of operating temperatures, longer life expectancy, and improved safety.[3-5] They are nowadays being used across a variety of areas, such as hybrid electric vehicles, electronic devices, and industrial equipment, etc.[6,7] According to the charge storage mechanisms, SCs can be classified into two categories: electrical double layer capacitors (EDLCs) and pseudocapacitors (PCs).[8] The EDLCs store the charges by fast ion adsorption/accumulation at the electrode/electrolyte interfaces, while the PCs are based on the reversible redox reactions, which can be realized by introducing redox species onto the electrode or into the electrolyte solution. To develop SCs with high performance (e.g., high capacitance), the choice or design of electrode materials plays the key role.

Conductive diamond, especially heavily boron-doped diamond (BDD) film, has been studied extensively in recent years as a promising electrode material for SC applications.[9-11] These applications are based on the unique physical, electrical and electrochemical properties of this p-type carbon material over other carbon allotropes, including its wide electrochemical potential windows in aqueous and non-aqueous solutions, easily changeable surface terminations or modifications, exceptional mechanical hardness, chemical resistance, and high stability in various harsh environments and at high voltage/current densities, etc.[12,13] Furthermore, various porous BDD nanostructures (e.g., diamond network,[14] diamond foam,[15] honeycomb diamond,[16] etc.) with enhanced surface areas can been fabricated using top-down, bottom-up, or template-/mask- free techniques. Additionally, BDD can been functionalized with different pseudo-capacitive species[9,17] or integrated with other sp^2 carbon materials[11,18] to form hybrid films. With these nanostructure based capacitor electrodes, SCs with improved performance have been achieved. The combination of BDD with redox-active species contained electrolytes has also been reported, leading to the construction of high-performance diamond PCs.[10,11]

Another kind of conductive diamond is n-type diamond doped with nitrogen or phosphorus dopant. Up to date, only limited studies deal with electrochemical properties and applications of n-type doped diamond,[19-23] although the nitrogen-doped diamond possesses a potential window similar to or even higher than that of BDD.[21-23] The electrochemical capacitance of a nitrogen-doped ultrananocrystalline diamond (N-UNCD) grown in a microwave plasma-assisted chemical vapor deposition (MWCVD) reactor was about 17 μF cm^{-2} in the phosphate buffer, which increased to more than 1 mF cm^{-2} after the treatment of the N-UNCD with low energy oxygen plasma for 3 h. The enhancement of the capacitance was explained by the enlargement of the surface area and the change of the hybrid sp^2/sp^3 carbon structure, originating from the introduction of oxygen functionalities and the etching of sp^2 bonded material in grain boundaries.[21] For phosphorus-doped diamond (PDD), a few studies have been reported on its electrochemical properties.[19,20] However, the utilization of PDD for SC applications has not been shown up to now. The possible reason is the semi-conducting property of PDD films, due to the

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insufficient doping of phosphorus (generally <10^{19} atoms cm^{-3}) in the PDD film. [24] This is because phosphorus is a larger atom compared to carbon. The incorporation of a high concentration of phosphorus atoms in the diamond lattice is relatively difficult. To some extent, the diamond lattice periodicity is disturbed, resulting in lattice defects and crystalline imperfection in the doped PDD films.[25,26] Moreover, phosphorus is known to exhibit a high activation energy (about 0.6 eV), another reason associated with the relatively low conductivity of as-grown PDD films.[27]

In this work, electrochemical properties of heavily phosphorus-doped nanocrystalline diamond (P-NCD) films with a doping level over 10^{20} cm^{-3}, grown using a MWCVD technique,[28] were investigated. To attain the P-NCD film with enhanced electrical conductivity, annealing treatment of as-grown P-NCD films was carried out in vacuum for hours. The potential of these as-grown and annealed P-NCD films as electrodes for the construction of SCs including EDLCs and PCs was then investigated. For EDLCs, 1.0 M Na_2SO_4 aqueous solutions were used. For PCs, 1.0 M Na_2SO_4 aqueous solutions containing 0.05 M $K_3Fe(CN)_6$/$K_4Fe(CN)_6$ redox electrolytes were applied.

Results and Discussion

Figure 1a shows the SEM image of as-prepared P-NCD film. The grain size was estimated to be smaller than 200 nm, while the preferential crystal orientation cannot be clearly observed. To check the possibility of utilizing this P-NCD films for electrochemical applications, its cyclic voltammograms (CVs) were recorded in different media, including acidic (1.0 M H_2SO_4), neutral (1.0 M Na_2SO_4), and alkaline (1.0 M NaOH) aqueous solutions. Figure 1b presents these CVs obtained at a scan rate of 100 mV s^{-1}. The scanned potential was ranged from -1.2 to 1.2 V. Taking a current density of ±0.01 mA cm^{-2} as the threshold, the obtained potential window of an as-grwon P-NCD film in aqueous solution is about 1.25 V, which is close to the value of the potential window for electrolytic water splitting (1.23 V), similar as that reported.[19] Compared to the potential window of a BDD electrode (about 3.2 V in aqueous solution),[12] the relatively narrow potential window of a P-NCD electrode might result from the high content of impurity and non-diamond carbon existed in the film. They can actually act as the electrocatalysts for water decomposition.[19] An anodic or cathodic pretreatment of the P-NCD films by applying potentials of 2.5 V and -2.5V (*vs*. Ag/AgCl) removed sp^2 carbons, as confirmed by *in situ* Raman spectroelectrochemistry, eventually resulting in a similar potential window as that for a BDD electrode. [19]

The as-grown P-NCD films were then utilized as an electrode for the construction of EDLCs. Their electrochemical performance was first investigated in 1.0 M Na_2SO_4 aqueous solution. Figure 2a illustrates the CVs of an as-grown P-NCD at different scan rates, ranging from 10 to 100 mV s^{-1}. At all scan rates, the obtained CVs show a quasi-rectangular shape, demonstrating a good electrical double layer capacitive characteristic of an EDLC. Through the integration of the charges during anodic and cathodic cycles, the calculated capacitance is 7.96, 8.77, 10.11, and 11.40 µF cm^{-2} at the scan rate of 100, 50, 20, and 10 mV s^{-1}, respectively.

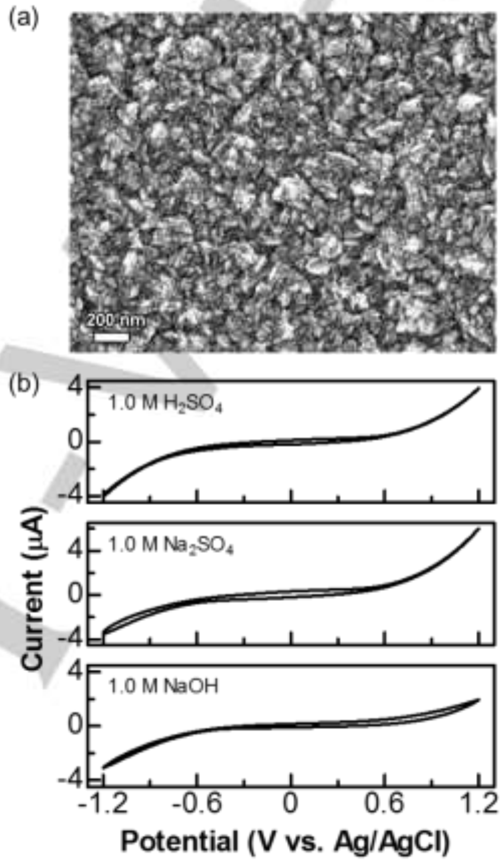

Figure 1. (a) SEM image of an as-grown P-NCD film; (b) CVs of the as-grown P-NCD film at a scan rate of 100 mV s^{-1} within a potential range of -1.2 – 1.2 V in 1.0 M H_2SO_4, 1.0 M Na_2SO_4, and 1.0 M NaOH.

The capacitance of the as-grown P-NCD films was further studied using galvanostatic charging/discharging (GCD) technique. Figure 2b presents the GCD curves at the current densities of 5, 2, and 1 µA cm^{-2}. All the acquired curves are almost symmetrical and linear in both charging and discharging parts, indicating the high reversibility and excellent capacitive behavior of this EDLC. The small *IR* drop observed in the discharging part is probably ascribed to the internal resistance of this as-grown P-NCD film. According to the charges during the charging/discharging cycles, the capacitance was estimated to be 5.60, 7.69, and 9.97 µF cm^{-2} at the current density of 5, 2, and 1 µA cm^{-2}, respectively. These values are comparable with those obtained using cyclic voltammetric technique. Interestingly, the capacitance of the as-grown P-NCD is slightly larger than that of a BDD electrode (3.6 – 7.0 µF cm^{-2} in 1.0 M Na_2SO_4).[9] The reason may be due to the presence of a high concentration of impurity or non-diamond carbon in the as-grown P-NCD films.

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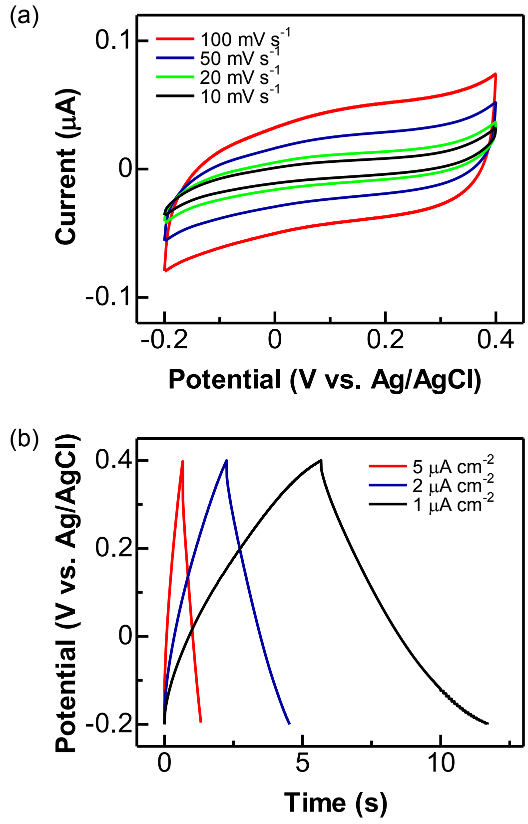

Figure 2. Performance of an as-grown P-NCD based EDLC in 1.0 M Na_2SO_4 at a potential window of -0.2 – 0.4 V: (a) CVs recorded at the scan rate of 100, 50, 20, and 10 mV s^{-1}; (b) GCD curves at the current density of 5, 2, and 1 μA cm^{-2}.

As a novel strategy to construct PCs, the addition of redox electrolytes into the solutions has been employed.[29,30] In this work, the capacitive behavior of as-grown P-NCD films was then investigated in 1.0 M Na_2SO_4 aqueous solutions containing 0.05 M $K_3Fe(CN)_6/K_4Fe(CN)_6$ redox electrolytes. These redox electrolytes were actually intensively studied on BDD and other sp^2 carbon electrodes, showing fast electron transfer and good reversibility. [10,11,31] Figure 3 shows the CV of an as-grown P-NCD film in 0.05 M $Fe(CN)_6^{3-/4-}$ at a scan rate of 100 mV s^{-1} within a potential range of -1.0 – 1.0 V, where the behavior of a n-type semiconductor material is revealed.[32] Note that, with a high phosphorus doping level (5×10^{20} cm^{-3}) of used P-NCD films, a high electrical conductivity is expected. Actually some dopant atoms segregated in the grain boundaries are generally not electrically active, they thus have no contribution to the conductivity of the film. This might be the origin of the semi-conducting property of these P-NCD films, although the measured doping level is actually higher than the charge carrier concentration.[24] More interestingly, the anodic and cathodic segments are fully overlapped. In other words, the values of anodic and cathodic currents at the same potentials are equivalent. However, no peaks related to the oxidation or reduction of $Fe(CN)_6^{3-/4-}$ have been observed in the anodic or cathodic segments. This might result from the following aspects: [12,20,33,34] i) the relatively high resistivity of the as-grown P-NCD film: because of the large activation energy of phosphorus donors ($E_{a,p}$ = 0.6 eV) and/or the relatively low concentration of charge carriers, the electron transfer through the P-NCD electrode/electrolyte interface is quite limited; ii) the influence of surface chemistry of the as-grown P-NCD films: surface terminations and the distribution of surface functional groups affect its wettability and even in some way block the access of redox molecules to approach it surface; iii) the existence of low amount of defect sites and non-diamond carbon impurity, leading to quite less amount of reactive sites. Note that, the electrochemical behavior of this as-grown P-NCD electrode differs from that of a quasi-metallic heavily BDD electrode, due to the different charge transfer mechanism at the electrode/electrolyte interfaces.[35] Surface state mediated charge transfer mechanism between the as-grown P-NCD film and the redox couples is more suitable for the explanation of the results obtained in Fig. 3.[36,37]

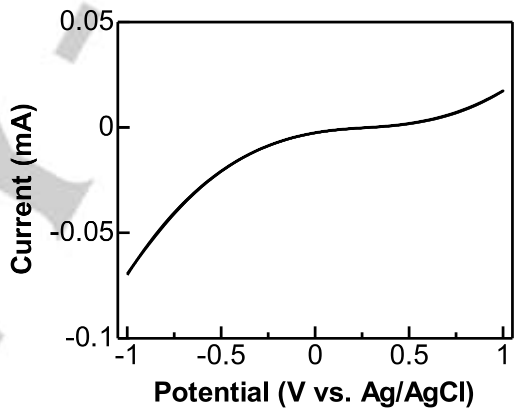

Figure 3. CV of an as-grown P-NCD film in 0.05 M $Fe(CN)_6^{3-/4-}$ + 1.0 M Na_2SO_4 at a scan rate of 100 mV s^{-1} in a potential window of -1 – 1 V.

Post-thermal treatment of these as-grown P-NCD films was then conducted in vacuum (about 5×10^{-2} mbar) at a temperature of 800 ºC for 7 h. In this way, the electronic as well as electrochemical properties of these as-grown P-NCD films are expected to be varied. The surface morphology of as annealed P-NCD film was checked using SEM. Figure 4a shows such a SEM image. Compared to that of an as-grown P-NCD film (Fig. 1a), virtually no changes of its surface morphology are observed after such thermal treatment. XPS spectra of these P-NCD films were further recorded to check the variation of their surface chemical states before and after such as annealing treatment. As shown in Fig. 4b and 4c, the recorded high-resolution C 1s spectra (black lines) for the as-grown and annealed P-NCD films have been de-convoluted into three and four peaks (colorful lines), respectively. For the as-grown P-NCD film, the peaks are centered at 288.3 eV (assigned to carboxylated C), 286.6 eV (assigned to hydroxylated C), and 284.8 eV (assigned to sp^3 hybridized C). For the thermally treated P-NCD film, the peaks at

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287.9 eV (assigned to carbonylated C), 286.5 eV (assigned to hydroxylated C), 285.3 eV (assigned to sp^3 hybridized C) and 284.4 eV (assigned to sp^2 hybridized C) are observed. Moreover, its peak corresponded to sp^2 carbon is much enhanced, compared to that for the as-grown P-NCD film. The atomic concentration of sp^2-hybridized carbon, calculated from this peak area, is about 51.34%, higher than that (33.56%) for sp^3-hybridized carbon. Hence a fraction of the sp^3-hybridized carbon has been converted into sp^2-hybridized carbon during this thermal annealing treatment. In addition, the atomic oxygen concentration of the as-grown P-NCD film was reduced from 25.22% to 20.68% after such thermal treatment, indicating a change of oxygen functionalized groups (e.g., types, amount, etc.) on the surface of the P-NCD film during the treatment. For example, the decomposition of carboxyl groups to carbonyl groups is clearly observed (green lines in Fig. 4b and 4c). Therefore, both the fraction conversion of the sp^3-hybridized carbon into sp^2-hybridized carbon and the changes of surface oxygen-functional groups during thermal annealing treatment lead to the increase of the electrical conductivity of P-NCD films. The surface wetting properties of these P-NCD films are varied as well. Eventually, the improved electrochemical activity of the annealed P-NCD films is expected to be observed in both inert and redox-active species contained solutions.

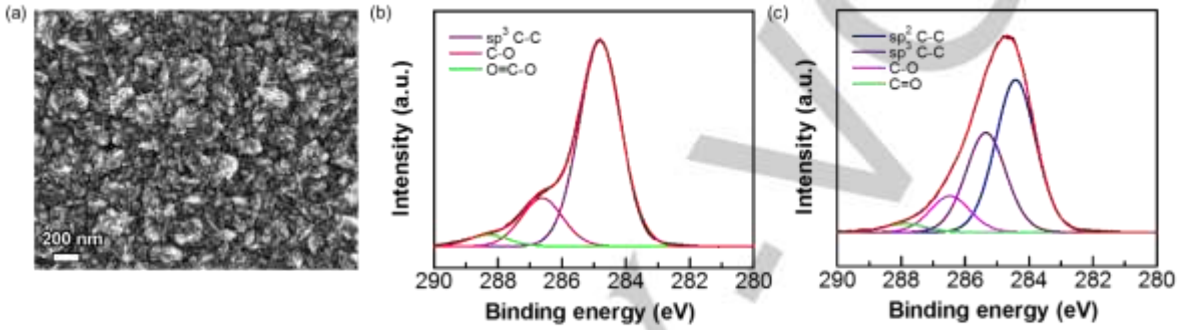

Figure 4. (a) SEM image of a P-NCD film after annealing treatment at 800 °C for 7 h; (b,c) C 1s XPS spectra of P-NCD films before (b) and after (c) such an annealing treatment. The black lines are experimental results and the colourful lines are simulated ones.

The annealed P-NCD film was then employed as the capacitor electrode for the construction of SCs. The related CVs (Fig. 5a) of the annealed P-NCD based EDLC in 1.0 M Na_2SO_4 was recorded in a potential window of -0.5 – 0.4 V at different scan rates. The slight deviation of the CVs from the rectangular shape of an ideal EDLC and the increase of the current at high potential regions are due to the generated over-potential of the electrode, the presence of impurity / non-diamond carbon in the film, and the variation of functional groups on the P-NCD surface. The calculated capacitance is 0.99, 1.24, 1.62, and 2.01 mF cm^{-2} at the scan rate of 100, 50, 20, and 10 mV s^{-1}, respectively. These values are almost 100 times higher, compared to the capacitances obtained from the as-grown P-NCD capacitor electrode.

The performance of a PC based on the annealed P-NCD film was studied in redox mediator contained aqueous solution (here 0.05 M $Fe(CN)_6^{3-/4-}$ in 1.0 M Na_2SO_4). As shown in Fig. 5b, CVs recorded in a potential window of -0.1 – 0.6 V at the scan rates of 100, 50, and 20 mV s^{-1} exhibit clearly both oxidation and reduction peaks of $Fe(CN)_6^{3-/4-}$. At all scan rates, the absolute values of anodic and cathodic peak currents are nearly identical. The separation between two peak potentials is relatively large, which also increases with an increase of the scan rates. These facts indicate an irreversible electrode process of $Fe(CN)_6^{3-/4-}$ on this annealed P-NCD film. According to these CVs, the capacitance of such a PC was estimated to be 24.05, 37.39, and 63.56 mF cm^{-2} at the scan rates of 100, 50, and 20 mV s^{-1}, respectively. These values are three orders of magnitude higher than those of as-grown P-NCD based EDLCs, and even higher than those of the reported BDD PCs using redox-active species contained electrolytes (e.g., 41.51 mF cm^{-2} at the scan rate of 10 mV s^{-1}).[10]

The variation of electrochemical characteristics of the P-NCD film in different electrolytes after annealing treatment may be mainly attributed to its improved electrical properties. To support such a statement, the charge transfer resistances of this P-NCD film before and after thermal treatment were measured by means of electrochemical spectroscopy impedance. As shown in Fig. 5c, the charge-transfer resistance of the annealed film, estimated from the diameter of the semicircle appeared in the related Nyquist plot, is much smaller than that for an as-grown P-NCD film (the inset of Fig. 5c). From our viewpoint, the following aspects are possible sources for such changes. First, high-temperature annealing results in the variation of the amount, distribution, and even the density of structural defects. The concentration of non-diamond carbon or impurity in the P-NCD films are also altered, eventually leading to improved electricity and then better electrochemical properties of P-NCD films. Second, the dynamic of the surface state or surface terminations (e.g., number of H atoms as well as number and type of oxygen functional groups) alters when such an annealing process is applied. This ends in the change of diamond surface electronic properties as well.[38] Furthermore, during the course of high-temperature treatment, some inactive phosphorus atoms located in interstitial and clustering sites or grain boundaries are movable and possible to be incorporated into substitutional sites, another positive fact for the improvement of electrical properties of these P-NCD films.[39,40] Once again, the charge transfer mechanism at the n-type diamond electrode and electrolyte interfaces is still poorly understood. More experiments and studies are further required.

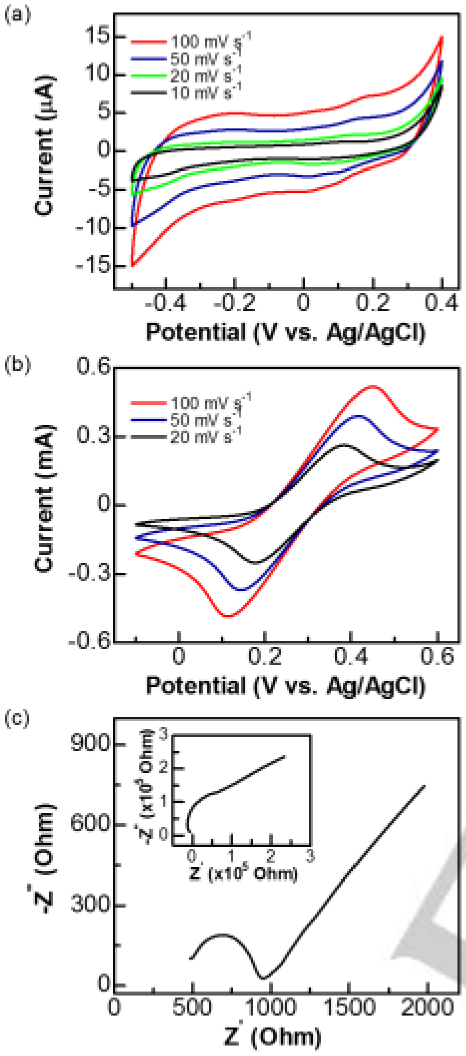

Figure 5. Performance of an annealed P-NCD film: CVs recorded at varied scan rates in the solutions of (a) 1.0 M Na_2SO_4 and (b) 0.05 M $Fe(CN)_6^{3-/4-}$ + 1.0 M Na_2SO_4; (c) Nyquist plot in 1.0 M Na_2SO_4 containing 0.05 M $Fe(CN)_6^{3-/4-}$. The inset shows the Nyquist plot of an as-grown P-NCD electrode in 1.0 M Na_2SO_4 containing 0.05 M $Fe(CN)_6^{3-/4-}$.

Conclusions

Heavily phosphorus-doped nanocrystalline diamond films have been grown by a plasma-enhanced chemical vapor deposition technique and further utilized for supercapacitor applications. By using an as-grown P-NCD film as the capacitor electrode, the formed EDLC possesses a capacitance of 11.40 μF cm^{-2} at a scan rate of 10 mV s^{-1}, slightly higher than that of a BDD EDLC. While in redox-active species contained electrolyte (e.g., 0.05 M $Fe(CN)_6^{3-/4-}$ + 1.0 M Na_2SO_4), semi-conducting behavior of the as-grown P-NCD has been observed (e.g., no peaks related to the oxidation or reduction reactions). This is mainly due to the relatively high resistance of the P-NCD film. Annealing treatment of this film at high temperatures in vacuum for hours did not change their surface morphology, but a higher conductivity has been acquired. Using this treated P-NCD films as the capacitor electrode, improved capacitance was achieved, for example, a EDLC capacitance of 2.01 mF cm^{-2} at a scan rate of 10 mV s^{-1} and a PC capacitance of 63.56 mF cm^{-2} at a scan rate of 20 mV s^{-1}. These values are 2-3 orders of magnitude higher than those obtained on as-grown P-NCD based EDLCs. The capacitance enhancement results from improved electrical conductivity and surface wettability, caused by the variation of surface termination, structural defects, and non-diamond carbon content during the course of annealing treatment of the as-grown P-NCD film. Moreover, the capacitance obtained on the annealed P-NCD based EDLCs and PCs are comparable or even higher than those of BDD based EDLCs and PCs, respectively. Therefore, PDD is a promising candidate to be applied as the electrode material for SC applications.

However, the basic electrochemical properties of PDD are actually not so clearly clarified up to now. The effects of the dopant concentration, electrical conductivity, crystallographic structure, non-diamond content, and surface chemistry on their electrochemical properties need to be clarified. For example, the comparison study of electrochemical properties of PDD films deposited on different substrates should be conducted especially when these films feature varied grain sizes, surface roughness, and amounts of non-diamond carbons. Tuning of electrical conductivity of PDD films with different approaches, such as post-thermal treatment in vacuum at high temperatures, co-doping with other kinds of dopants (e.g., boron), and post-electrochemical treatment at high anodic/cathodic electrochemical current/potentials must be conducted. Regarding the applications of PDD for the SC construction, more detailed investigations on the performance of P-NCD based SCs (e.g., cycling stability, resistance, power and energy densities, etc.) are still required using various techniques, including cyclic voltammetry, the GCD technique and impedance spectroscopy, these results demonstrate.

Experimental Section

The P-NCD films were grown on molybdenum substrates by a plasma-enhanced chemical vapor deposition (CVD) technique.[28] Prior to film deposition, the molybdenum substrate was seeded with a nano-diamond (diameter: 3 – 6 nm) colloidal solution using a sonic wave bath. For the growth of P-NCD films, the applied parameters were as following: microwave power of 750 W, growth temperature of 900 °C, chamber pressure of 25 Torr, and growth time of 4 hour. The thickness of as-grown P-NCD films was about 250 nm. The main precursors were methane and phosphine, carried by hydrogen gas. During the growth, the CH_4/H_2 flow ratio was kept at 1 %, while the P/C ratio was 2. The phosphorus concentration in the doped diamond film, estimated using secondary ion mass spectrometry in the same way as reported,[28] was 5×10^{20} cm^{-3}. As-grown P-NCD films were further annealed in a thermal chemical vapor deposition device. The applied annealing conditions were as following: annealing temperature of 800 °C, pressure of about 5×10^{-2} mbar, and annealing time of 7 h.

The surface morphologies of these P-NCD films were investigated with a field emission scanning electron microscopy (FESEM, Zeiss ultra 55, Germany). Their surface chemical states were characterized using X-ray photoelectron spectroscopy (XPS, Surface Science Instruments, SSX-100 S-probe photoelectron spectrometer, USA) with an Al Kα radiation of 200 W. Survey spectra were firstly recorded in a range of 0 to 1200 eV with a resolution of 1 eV and a spot size of 800 μm². The resolution for the high-resolution scans was set to 0.1 eV. The obtained XPS spectra were analysed using CasaXPS processing software (version 2.3.16 PR 1.6) and the binding energy were referenced to the C 1s signal (284.8 eV).

Electrochemical measurements were carried out on a CHI660E Potentiostat/Galvanostat (Shanghai Chenhua Inc., China) using a standard three-electrode cell, where the P-NCD film acted as the working electrode, an Ag/AgCl (3M KCl) as the reference electrode, and a coiled Pt wire as the counter electrode. The effective area of the working electrode that exposed to the electrolyte was a circular area of approximately 0.05 cm^{-2}. For the construction of diamond EDLCs and PCs, 1.0 M Na_2SO_4 and 1.0 M Na_2SO_4 containing 0.05 M $K_3Fe(CN)_6/K_4Fe(CN)_6$ aqueous solutions were utilized, respectively. Their performance was examined using cyclic voltammetry at different scan rates, the galvanostatic charging/discharging method at different current densities, and electrochemical impedance spectroscopy technique at open circuit potentials in the frequency range of 0.01 to 10^6 Hz.

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Keywords: diamond electrochemistry • supercapacitor • phosphorus-doped nanocrystalline diamond • capacitance • electrical conductivity

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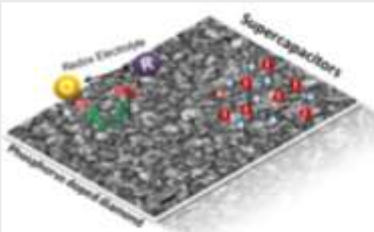

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